DataScientest

Anomaly Detection in Industrial Components

Relevant cursus: DS

Difficulty Level: 10/10

Description of the project:

The objective of this project is to develop a model capable of detecting anomalies in an industrial part using an image. To achieve this, a convolutional neural network (CNN) or any other architecture suitable for this type of problem can be used. Several approaches can be considered for this project, with either supervised or unsupervised learning.

The first objective is to perform binary classification on the dataset images to isolate those presenting an anomaly. The second objective is to identify the detected anomaly, thus transforming the problem into a multi-class classification task. Each part is associated with a series of specific anomalies.

Resources to consult:

- **Data:**
  - MVTec Industrial 3D Object Detection Dataset (MVTec ITODD): https://www.mvtec.com/company/research/datasets/mvtec-itodd
  - RAD: A Comprehensive Dataset for Benchmarking the Robustness of Image Anomaly Detection: https://github.com/hustCYQ/RAD-dataset
- **Bibliography:**
  - The MVTec 3D-AD Dataset for Unsupervised 3D Anomaly Detection
  - AUPIMO: Redefining Visual Anomaly Detection

Project validation conditions:

- A **report** on data exploration, visualization, and preprocessing
- A modeling **report**
- A **final report** and the associated **GitHub** repository